To: Dr. Naqaa Abbas, ENGL 220 - Technical Writing - Spring 2026
From: Team Cocktail—Aazim Rafeeq, Hiba Syed, Kel Doliente, Khaldoun Saadeddine, and Youssef Abou Ali
Date: Sunday, April 26, 2026
Subject: Technical Report - Lack of Pedestrian Walkability in Education City, Qatar

TECHNICAL REPORT

I. Abstract

Education City (EC) in Qatar spans 12 square kilometers and hosts several world-class universities, research institutions, and cultural landmarks. Despite EC's design as a walkable academic hub, most of its population relies on private vehicles for short-distance travel due to high temperatures. Our team evaluated four potential solutions: The Park and Plant tree-planting initiative, cool pavement coatings, climate-controlled enclosed walkways, and solar-powered misting systems. A weighted decision matrix was used across four criteria: thermal comfort, sustainability, coverage and scalability, and feasibility. Park and Plant achieved the highest score, offering natural cooling through native tree shade, and campus-wide scalability without new infrastructure. The recommendation aligns with QNV 2030's pillars of environmental sustainability and human development, leveraging native tree species such as Sidra and Ghaf to provide natural shade, improve air quality, sequester carbon, and create a more walkable campus environment.

II. Introduction

Education City (EC) is a 12-square-kilometer Qatar Foundation campus in Doha designed as a walkable academic hub. It hosts Hamad Bin Khalifa University, Texas A&M University at Qatar, and Carnegie Mellon University in Qatar, alongside research centers, Oxygen Park, the Qatar National Library, and the Qatar Foundation headquarters, all connected by a 24-station tram system and pedestrian network. Despite this design intent, only 17% of students, staff, and visitors walk between establishments, while 57% rely on private vehicles (Al-Mohannadi et al., 2023). This dependency drives transport-sector emissions, in direct conflict with the Third National Development Strategy's target of a 25% reduction in greenhouse gas emissions by 2030 (National Planning Council, 2024), while also reducing physical activity and weakening cross-campus community life.

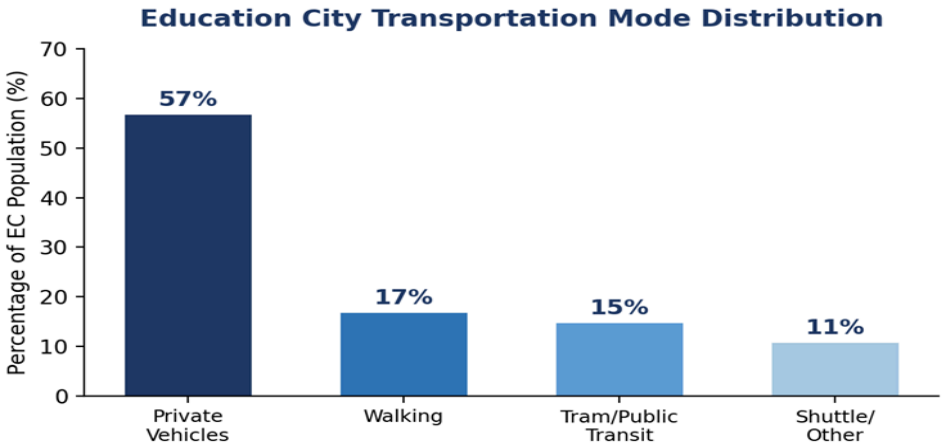

Figure 1: *Transportation mode distribution among the Education City population (Al-Mohannadi et al., 2023).*

The primary barrier is climate. Summer daytime temperatures average 45°C with humidity between 40% and 60%, making sustained outdoor walking uncomfortable and often unsafe (Visit Qatar, 2026). Qatar's National Master Plan responds to this challenge by recommending 60–70% shade coverage on pedestrian walkways in hot-arid conditions (Ministry of Municipality and Environment, n.d.) — a benchmark EC's current infrastructure does not meet.

This report investigates the causes of poor walkability in Education City, evaluates four potential solutions using a weighted decision matrix, and suggests the most effective and practical option for the campus environment.

III. Methodology

This study used a mix of different methods to get a well-rounded understanding of our topic. It included a structured literature review and an engineering-based evaluation framework.

1. Literature Review

The literature review was performed using databases like Google Scholar, Scopus, and the HBKU library. The search terms used were “walkability in hot climates," “pedestrian thermal comfort in Qatar," and “effective shaded walkways," with a preference for the studies that were performed in the Gulf region. The most relevant sources were Al-Mohannadi et al. (2023) for insights on urban mobility and transit-oriented development, Middel et al. (2023) and Wang et al. (2024) to get data on cooling strategies, and the Third Qatar National Development Strategy (National Planning Council, 2024) for understanding the bigger picture of policy. A total of 15 to 20 academic and institutional sources were reviewed.

2. Institutional and Contextual Analysis

Beyond the literature, the team also looked at what is already happening in Education City. Initiatives such as the Park and Plant program and the Quranic Botanic Garden provided useful examples of how sustainable landscaping is already being used locally to manage heat. The team also spent many weeks informally collecting data regarding the issue of walkability and thermal comfort. This helped make sure that the analysis stayed connected to how the space is actually experienced on a day-to-day basis.

3. Engineering Evaluation Tools

For the comparison of all the possible solutions in a structured manner, a weighted decision matrix was used to rank the solutions for more accuracy. Criteria such as thermal comfort (30%), sustainability (30%), coverage and scalability (25%), and feasibility (15%) were ranked. Each solution was then scored on a 1-5 scale, with the existing EC canopies being used as the reference. The final ranking was then calculated by multiplying each score by its weight and adding all the results up. Using this method helped the team make an accurate decision instead of just depending on assumptions and personal opinions.

IV. Results

Our research confirmed that Education City's walkability problem is driven by a combination of extreme heat, insufficient shade, and a campus layout that favors car use over walking. With temperatures regularly exceeding 40°C from June through September and humidity levels between 40% and 60%, walking between buildings becomes physically exhausting. Only 17% of EC's population walks between facilities, while 57% prefer using their own cars (Al-Mohannadi et al., 2023). Although the campus tram serves 24 stations, many walkway segments between tram stops and building entrances remain unshaded, limiting its usefulness as a pedestrian mobility tool (Luomi et al., 2014). Gulf countries have historically been designed around cars, with wide roads and minimal space designed for walking, and Education City reflects this pattern despite its original vision as a walkable campus.

After reviewing these circumstances, our team identified four potential solutions. These fall into two broad approaches: nature-based cooling and engineered cooling systems.

Solution 1: Park and Plant Initiative

Park and Plant is Qatar Foundation's existing tree-planting program, which aims to line EC walkways with native species such as Sidra and Ghaf trees to create continuous shaded corridors. These trees block direct solar radiation while still allowing airflow, which research has shown to be more effective than full coverage. Al Tahir et al. (2025) found that 75% canopy shading provides better airflow and cooling than complete coverage. The initiative started with 2,500 trees and has a target of 20,000, supported by community engagement programs like "Adopt a Tree" and corporate sponsorship through "Green Lungs" (Qatar Foundation, n.d.). A comparable study conducted on a university campus in Dubai showed that planting Ghaf and Acacia trees reduced the annual number of hours classified as "warm/hot" from 1,291 to fewer than 300 (Naboni et al., 2018). However, native trees require several years to reach full canopy maturity, and irrigation and seasonal pruning represent ongoing maintenance needs.

Figure 2: *Ghaf Tree in arid climate (Plant 2018)*

Solution 2: Cool Pavement Coatings

Cool pavement coatings are reflective layers applied directly over existing walkway surfaces to increase their surface reflectivity, reducing the amount of solar heat the pavement absorbs and radiates back to pedestrians. Studies conducted in Phoenix showed that these coatings can lower surface temperatures by 2.6–4.9°C compared to conventional materials used for surfaces (Middel et al., 2023). The coatings are relatively affordable and can be applied with minimal construction, making them one of the easiest solutions to implement. However, there is a significant trade-off: reflective pavements may increase mean radiant temperature at the pedestrian level by 2–3°C, meaning walkers could feel hotter despite the ground being cooler (Middel et al., 2023). The coatings also degrade over time due to UV light exposure, dust, and foot traffic, requiring periodic reapplication (Bloomberg Associates, 2019). Research further suggests that cool pavements are most effective when paired with overhead shade, but combining both may not be cost-effective (Ghoniem et al., 2025).

Figure 3: *Cool pavement coating application (WHYY 2019)*

Solution 3: Climate-Controlled Enclosed Walkways

Enclosed and air-conditioned walkways provide complete thermal protection regardless of the surrounding conditions, offering the strongest thermal comfort of any solution and making them especially helpful for vulnerable groups such as children, the elderly, and migrant workers who may lack access to private vehicles. Qatar already has working previous examples of this approach, including an air-conditioned pedestrian bridge in the Industrial Area (Qatar Living, 2015) and the elevated enclosed bridge connecting the Doha metro station to the Mall of Qatar. However, the energy demands of cooling large enclosed structures in a climate where outside

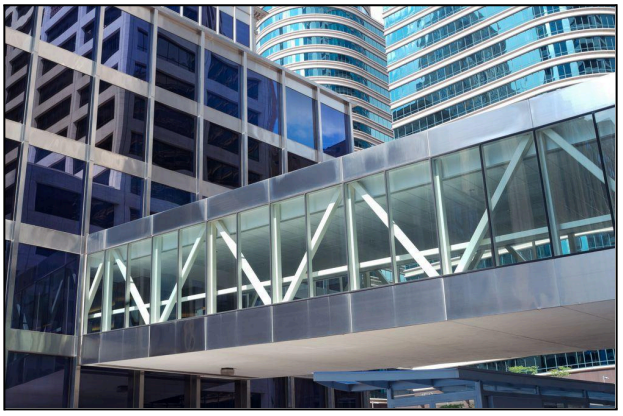

Figure 4: *External view of an enclosed walkway connecting urban commercial buildings (Treehugger/Joe Ferrer).*

temperatures exceed 45°C are substantial, likely contradicting the sustainability goals unless powered totally by renewable energy. In addition, effective enclosed networks must connect heavily populated locations to justify their high cost, a condition that Education City's spread-out layout does not meet. Construction costs are high, and enclosed environments may also reduce the social spirit associated with open public spaces (Litman, 2023).

Solution 4: Solar-Powered Misting Systems

This solution suggests replacing existing EC canopy structures with high-pressure misting nozzles powered by rooftop solar power panels, which produce tiny droplets that evaporate quickly, absorbing heat from the surrounding air and lowering pedestrian-level temperatures. Smart thermostatic sensors would activate the system only when the temperature exceeds a set threshold, conserving both water and energy. Research by Wang et al. (2024) shows that shade combined with misting reduces the global thermal climate index by an average of 12.3°C, a much greater impact than shade alone, which reduces it by 3.4°C. Solar-powered aerial misting has been used in similar GCC outdoor environments in Saudi Arabia and at Expo 2020 Dubai, proving its regional practicality (ResearchGate, 2013). However, misting effectiveness decreases greatly at humidity levels above 60–70% (Wang et al., 2024), and Qatar's summer humidity levels often reach 40–60%, approaching the acceptable limit. Coverage is also limited to existing canopy areas, leaving exposed walkway sections untreated. The nozzles require regular maintenance to prevent clogs and jams in Qatar's rough water conditions.

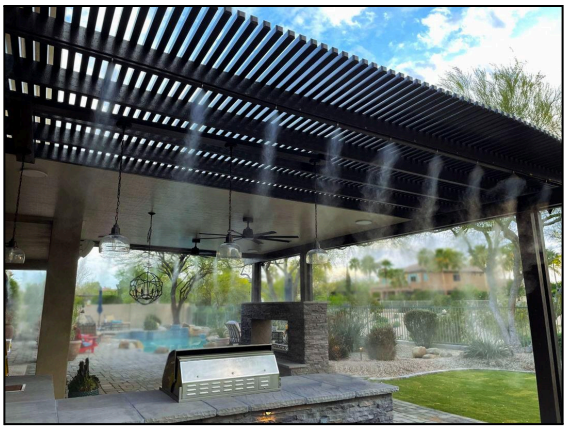

Figure 5: *Misting system on a patio (Solartech).*

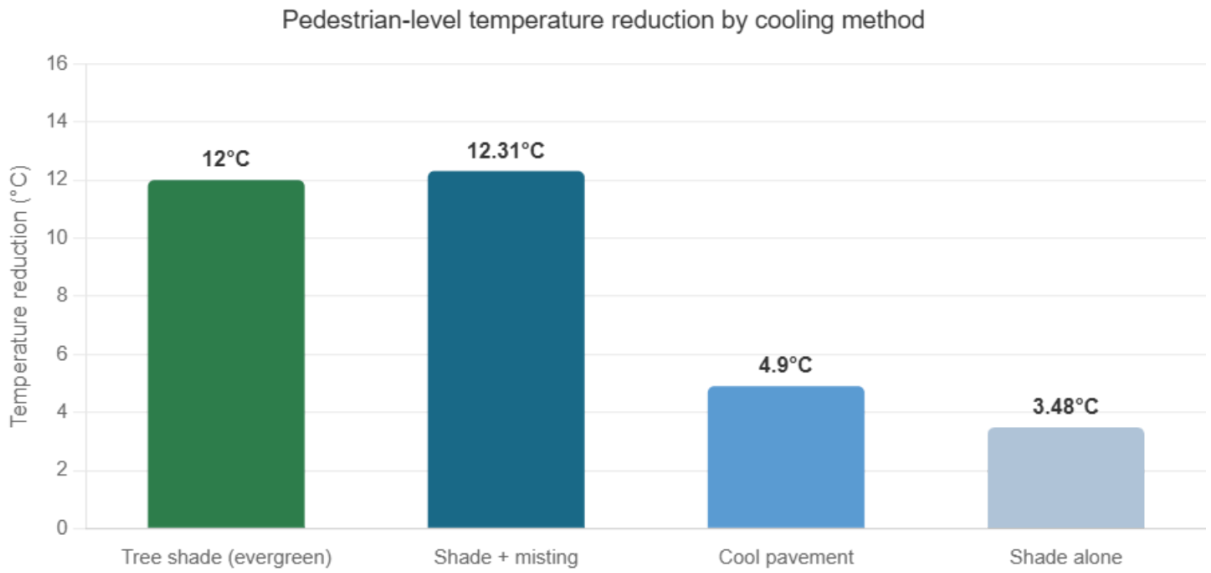

Figure 6: *Pedestrian-level temperature reduction by cooling method, compiled from multiple sources (Communications Earth & Environment, 2024; Middel et al., 2023; Wang et al., 2024)*

V. Analysis

The weights assigned to each criterion were determined based on their importance in QNV 2030 and logical reasoning. For instance, thermal comfort was given greater weight due to its relevance to the problem being assessed, while sustainability was assigned equal importance due to its significance in NDS3 (National Planning Council, 2024). The prevailing canopies around EC were selected as the reference concept, and solutions were evaluated on a 1–5 scale relative to this benchmark, supported by the cited literature. A score of 1 indicates a solution significantly worse than the reference, a 3 indicates equal performance, and a 5 indicates a significantly better outcome (American Society for Quality, 2026). Each rating was then multiplied by its corresponding percentage weight, and the resulting weighted scores were summed. The concept with the highest total score was ranked first. Based on this analysis, the Park and Plant solution was identified as the most effective.

Concepts											
Selection Criteria	**Weight**	**Park and Plant**		**Cooling and Reflective Coatings**		**Closed Air-conditioned Walkways**		**Solar-Powered Misting Systems**		**EC Canopies (Reference)**	
Customer Needs	**Σ = 100%**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**
Thermal Comfort	**30%**	3	0.9	1	0.30	5	1.50	4	1.20	3	0.90
Sustainability	**30%**	5	1.5	2	0.60	1	0.30	4	1.20	3	0.90
Coverage & Scalability	**25%**	4	1.00	5	1.25	2	0.50	3	0.75	3	0.75
Feasibility	**15%**	4	0.60	4	0.60	1	0.15	3	0.45	3	0.45
Total Score		4.00		2.75		2.45		3.60		3.00	

Table 1: *Weighted Decision Matrix for Concept Scoring*

The comparison of the thermal comfort performance among the solutions highlights their varying capacity to reduce pedestrian-level heat exposure. Park and Plant scores a 3 (equal to reference) because while mature tree canopy can reduce temperatures by up to 12°C (Communications Earth & Environment, 2024), newly planted trees require 3–5 years to establish meaningful shade coverage, meaning immediate cooling matches the current canopy infrastructure. In comparison, enclosed walkways score the highest at 5 because they provide complete climate control regardless of external conditions (Litman, 2023). Solar-powered misting systems score a 4, as shade combined with misting reduces the Universal Thermal Climate Index by an average of 12.31°C, though effectiveness declines at humidity above 60–70% (Wang et al., 2024). Cool pavement coatings score the lowest at 1, because while they lower surface temperatures by 2.6–4.9°C, they may actually increase mean radiant temperature at the pedestrian level by 2–3°C, potentially worsening thermal discomfort for walkers (Middel et al., 2023).

As for sustainability, Park and Plant score the highest at 5 because trees actively sequester carbon dioxide, filter particulate matter, reduce the urban heat island effect, and improve

biodiversity—directly supporting NDS3's target of 25% greenhouse gas reduction by 2030 (National Planning Council, 2024). Unlike engineered systems that consume energy, trees generate compounding environmental benefits as they mature. Solar misting scores a 4 because its solar-powered operation produces zero operational emissions, though it still consumes water—a scarce resource in Qatar (ResearchGate, 2013). Cool pavement coatings score a 2 because, while they reduce surface heat, they offer no ecological benefit and require periodic chemical reapplication that generates waste (Bloomberg Associates, 2019). Enclosed walkways score the lowest at 1, as maintaining below-ambient temperatures in large enclosed structures using air-conditioning systems requires significant energy consumption, directly conflicting with sustainability goals unless powered entirely by renewable sources (Litman, 2023).

For coverage and scalability, cool pavement coatings score the highest at 5 because they can be applied rapidly to any existing pavement surface across the entire campus with minimal structural modification (Bloomberg Associates, 2019). Park and Plant scores a 4 because tree planting can be phased across the full pedestrian network. The initiative already has a target of 20,000 trees and community programs like "Adopt a Tree" and "Green Lungs" that enable continuous expansion (Qatar Foundation, n.d.). Solar misting scores a 3 (equal to reference) because coverage is limited to existing canopy footprints, leaving uncovered walkway segments unaddressed without further canopy extensions. Enclosed walkways score the lowest at 2 because effective networks require connecting densely populated locations (Litman, 2023), and Education City's dispersed layout makes campus-wide enclosure impractical and prohibitively expensive.

Finally, for feasibility, Park and Plant and cool pavement coatings both score a 4 because they can be implemented within Education City's existing infrastructure without major construction. Park and Plant holds a decisive advantage here: it is an already operational Qatar Foundation initiative with established staffing, budgets, governance, and community engagement programs; no new procurement processes or stakeholder agreements are required (Qatar Foundation, n.d.). CMU-Q has already planted 109 trees, and Qatar Chamber has formally endorsed the initiative. Solar misting scores a 3 because, while it retrofits existing canopy structures, it introduces new maintenance complexities, including nozzle blockages from Qatar's hard water and concerns about bacterial buildup. Enclosed walkways score the lowest at 1, as they require heavy construction, high capital investment, and ongoing energy costs that are difficult to justify for an open campus layout (Litman, 2023).

After tallying up the scores on the decision matrix in Table 1, Park and Plant scored the highest at 4.00 points. It excels in sustainability and performs strongly across all other criteria, making it the most balanced and context-appropriate solution. Solar misting scored second at 3.60, performing well in thermal comfort and sustainability but limited by coverage and maintenance concerns. The EC canopy reference scored 3.00, as intended. Cool pavement coatings scored 2.75, excelling in coverage but critically weak in thermal comfort. Enclosed walkways, with a score of 2.45, achieve the best thermal comfort but are severely hindered by poor sustainability, limited scalability, and high costs.

However, a few caveats are worth noting. The weights reflect team judgment informed by QNV 2030 priorities, so alternative weightings (placing feasibility above sustainability, for instance) could shift the ranking. The scoring is also compensatory: Park and Plant's high sustainability score partly masks its weakest dimension, immediate thermal comfort, where newly planted trees cannot match the other engineered alternatives for 3–5 years, leaving pedestrians without relief for approximately 4-5 summers. The matrix also assumes stable implementation conditions and does not account for risks

such as loss of tree life during extreme heat events, rising water costs if Qatar's groundwater restrictions tighten, or slower-than-expected community uptake of the "Adopt a Tree" and "Green Lungs" initiatives. These risks do not overturn the ranking, but they clarify that Park and Plant's advantage is conditional on sustained institutional commitment and should be interpreted as the best balanced option rather than a risk-free one.

VI. Discussion of Recommendation

After evaluating all four solutions using the weighted decision matrix, the research confirms that expanding Qatar Foundation's Park and Plant initiative is the most effective, sustainable, and feasible response to Education City's walkability challenge. Park and Plant achieved the highest weighted score (4.00/5.0). Despite this, it also has its implementation risks, factors affecting implementation speed, and tradeoffs.

Proven and Already Operational

This existing framework eliminates the need for new governance structures or procurement processes, giving it a feasibility advantage that no other solution offers. CMU-Q has contributed to this initiative and has been formally endorsed by the Qatar Chamber. Over 2,500 trees have already been planted with a target of 20,000, supported by "Adopt a Tree" and "Green Lungs" sponsorship programs (Qatar Foundation, 2026).

Evidence of Cooling Effectiveness

Tree shade reduces pedestrian temperatures by up to 12°C, effective in arid climates (Communications Earth & Environment, 2024). On a Dubai campus, Ghaf and Acacia planting cut “warm/hot” hours by over 75% (Naboni et al., 2018), while 75% canopy coverage optimizes airflow and cooling (Al Tahir et al., 2025). Strategic tree planting significantly improves outdoor comfort and supports walking in hot-arid environments (Baobeid et al., 2021).

Promotes Walkability

Shaded pathways in Education City promote walking, reducing reliance on carbon-intensive transport while enhancing green space interaction. This supports Qatar’s sustainability goals (Patriche et al., 2023) and serves as an effective cooling strategy that helps lower vehicle emissions, which reached 34,239.15 tCO_2e in 2024 (Qatar University, 2024).

Native Species and Implementation Plan

Sidra and Ghaf are drought-tolerant, low-water species suited to Qatar’s climate. The Qur'anic Botanic Garden at Education City maintains over 6,800 native plants for planting (QBG, n.d.). Implementation occurs in three phases: Phase 1 (Years 1–2) focuses on high-traffic corridors linking tram stations to key buildings; Phase 2 (Years 2–4) expands to housing, parking, and Oxygen Park; Phase 3 (Years 4–6) achieves 60–70% campus-wide shade coverage, measured through pedestrian counts and comfort surveys. Scaling can be accelerated through partnerships with national tree-planting initiatives managed by Ashghal using treated wastewater irrigation (Hamad Bin Khalifa University, 2025; Ataullah, 2023).

Alignment with QNV 2030

Park and Plant directly supports NDS3's targets: 25% greenhouse gas reduction, sustainable transportation, and increased urban green space (National Planning Council, 2024). Trees sequester carbon, reduce the urban heat island effect, improve air quality, and create walkable environments

that decrease car dependency. Qatar's National Master Plan recommends 60–70% pedestrian shade coverage (Ministry of Municipality and Environment, n.d.), a goal that tree planting directly achieves.

Risks

The recommended initiative depends on partnerships with Ashghal for treated wastewater irrigation and on Qatar Foundation funding for staffing, manpower, and community collaboration. Any decline in these inputs would directly reduce efficiency and hinder progress toward the target (Ataullah, 2023; Qatar Foundation, n.d.). Additionally, although Sidra and Ghaf trees are drought-tolerant and well-adapted to Qatar's climate, they are vulnerable during their early growth stages (0–5 years). This increases the risk of mortality, requiring temporary shade and careful irrigation management.

Factors Affecting Implementation Speed

A key factor affecting the implementation speed of the Park and Plant initiative is Qatar's extreme climate, which increases mortality rates among young trees (years 0-5) that are vulnerable to drying and therefore require consistent care (Communications Earth & Environment, 2024). Another constraint is the availability of treated wastewater from Ashghal for irrigation (Ataullah, 2023). Finally, adequate staffing is required to maintain the trees throughout their growth. This depends on the previously mentioned factors, as well as the Qatar Foundation's budget allocation and planning for landscape maintenance (Qatar Foundation, n.d.).

Trade-offs

Sidra and Ghaf trees provide optimal benefits only after reaching full maturity, which takes around five years. In practical terms, this means that the cohort entering EC in 2026 will graduate before the canopy reaches functional shade. As a result, pedestrians will not immediately experience the significant cooling effects these trees offer in their early stages, compared to engineered solutions that provide instant thermal relief (Communications Earth & Environment, 2024). Choosing an environmentally conscious approach reduces reliance on carbon-emitting systems and improves air and environmental quality, supporting NDS3's greenhouse emissions target (Qatar University, 2024; National Planning Council, 2024).

References:

Al-Mohannadi, M., Awwaad, R., Furlan, R., Grosvald, M., Al-Matwi, R., & Isaifan, R. J. (2023). Sustainable status assessment of the transit-oriented development in Doha's Education City. Sustainability, 15(3), 1913. https://doi.org/10.3390/su15031913

Al Tahir, I., Abd El Fattah, A., Mohammed, M., Asif, M., & Almahdy, O. (2025). Evaluating the performance of outdoor shading devices on human thermal comfort in hot humid climates: A case study of Dhahran. Building and Environment, 271, 112625. https://doi.org/10.1016/j.buildenv.2025.112625

Ali, A.S.A., (2012). Ghaf tree. PlantAli.Blogspot. https://share.google/14TSvSkGjK6KcZdhN

Ali, S. B., Patnaik, S., & Madguni, O. (2023). Measures for improved pedestrian walkability in urban areas: A review. IOP Conference Series: Earth and Environmental Science, 1130, 012025. https://doi.org/10.1088/1755-1315/1130/1/012025

American Society for Quality. (2026). Decision matrix. https://asq.org/quality-resources/decision-matrix

Ataullah, S. (2023, January 8). Smart irrigation solution helps increase public parks' green areas manifolds. The Peninsula Qatar. https://thepeninsulaqatar.com/article/08/01/2023/smart-irrigation-solution-helps-increase-public-parks-green-areas-manifolds

Baobeid, A., Koç, M., & Al-Ghamdi, S. G. (2021). Walkability and its relationships with health, sustainability, and livability: Elements of physical environment and evaluation frameworks. Frontiers in Built Environment, 7, 721218. https://doi.org/10.3389/fbuil.2021.721218

Bloomberg Associates. (2019). Mitigating urban heat island effects: Cool pavement interventions. https://www.bbhub.io/dotorg/sites/32/2019/08/20190516_Cool-Pavement-Research-FINAL.pdf

Communications Earth & Environment. (2024). Cooling efficacy of trees across cities is determined by background climate, urban morphology, and tree trait. Nature, 5, 1908. https://doi.org/10.1038/s43247-024-01908-4

Environment Agency – Abu Dhabi. (n.d.). Ghaf tree (Prosopis cineraria). https://www.ead.gov.ae/en/discover-our-biodiversity/plants/ghaf-tree

Ghoniem, M., Fahmy, A., & Kamel, T. M. (2025). Evaluating urban surface materials: Albedo optimization and shading for pedestrian thermal comfort in hot-arid environments. Civil Engineering and Architecture, 13(6), 4331–4352. https://doi.org/10.13189/cea.2025.130616

Hamad Bin Khalifa University. (2025). QBG distributes 5000 native trees. https://www.hbku.edu.qa/en/news/qbg-distributes-5000-native-trees

iStock. (2015). Office Towers Reflections and Skyway stock photo. iStock. https://www.istockphoto.com/photo/office-towers-reflections-and-skyway-gm481742660-69578073

Litman, T. (2023). Cool walkability planning: Providing pedestrian thermal comfort in hot climate cities. Journal of Civil Engineering and Environmental Sciences, 9, 79–86. https://doi.org/10.17352/2455-488X.000073

Luomi, M., Crist, J. T., Alam, B., & Shakir, M. B. (2014). Environmental sustainability in Qatar's Education City: Strategies, initiatives, and education. QScience Connect, 2013(1), 41. https://doi.org/10.5339/connect.2013.41

Middel, A., Schneider, F. A., Ortiz, J. C., Vanos, J. K., & Sailor, D. J. (2023). Evidence-based guidance on reflective pavement for urban heat mitigation in Arizona. Nature Communications, 14(1). https://doi.org/10.1038/s41467-023-36972-5

Ministry of Municipality and Environment. (n.d.). Qatar National Master Plan: Climate Change Adaptation and Mitigation – Stage 4: Strategy Report. https://www.mme.gov.qa/QatarMasterPlan/Downloads-qnmp/ClimateChange/Stage%204%20-%20Strategy%20Development/Stage%2004%20Strategy%20Report%20English_s.pdf

Naboni, E., Desert, A., & Garcia, D. (2018). Thermal comfort maps to estimate the impact of urban greening on the outdoor human comfort. Urban Forestry & Urban Greening, 35, 12–25. https://doi.org/10.1016/j.ufug.2018.08.013

National Planning Council. (2024). Third Qatar National Development Strategy 2024–2030. https://www.npc.qa/en/planning/nds3/Documents/QNDS3_EN.pdf

Patriche, D., Marin, M., & Cuculici, R. (2023). Walkability and sustainable urban planning in the context of climate change. Sustainability, 15(18), 13813. https://doi.org/10.3390/su151813813

Qatar Foundation. (n.d.). Park and Plant. Retrieved March 13, 2026, from https://www.qf.org.qa/park-and-plant

Qatar Living. (2015, June 10). Qatar's first air-conditioned pedestrian bridge. https://www.qatarliving.com/en/article/qatars-first-air-conditioned-pedestrian-bridge

Qatar University. (2024). QU carbon footprint report 2024. https://www.qu.edu.qa/en-us/Offices/FacilitiesGeneralServices/Documents/QU%20carbon%20footprint%20report%202024%20final.pdf

Qur'anic Botanic Garden. (n.d.). About us. https://qbg.org.qa/about-us/

ResearchGate. (2013). Solar energy powering up aerial misting systems for cooling surroundings in Saudi Arabia. https://www.researchgate.net/publication/257051640

Schmidt, S., (2025). Philadelphia tests out 'cool pavement' in Hunting Park to combat summer heat. PlanPhilly. https://share.google/j8cllTehR4f89F6kh

SolarTech. (2026). Outdoor Misting Systems For Cooler Backyards. SolarTech. https://solartechonline.com/misting-systems/

Visit Qatar. (n.d.). Qatar's climate. Retrieved March 7, 2026, from https://visitqatar.com/intl-en/about-qatar/climate#Summer

Wang, Z., Li, Y., & Chen, Q. (2024). Mist spraying as an outdoor cooling spot in hot-humid areas: Effect of ambient environment and impact on short-term thermal perception. Buildings, 14(2), 336. https://doi.org/10.3390/buildings14020336